Uses of Prime Factor Decomposition

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Uses of prime factor decomposition

How can I use PFD to identify a square or cube number?

- If all the **indices** in the prime factor decomposition of a number are **even**, then that number is a **square** number
  - E.g. The prime factor decomposition of 7056 is $2^4 \times 3^2 \times 7^2$
  - All powers are even so it must be a square number
    - It can be written as $(2^2 \times 3 \times 7)^2$
- If all the indices in the prime factor decomposition of a number are **multiples of 3**, then that number is a **cube** number
  - E.g. The prime factor decomposition of 1728000 is $2^9 \times 3^3 \times 5^3$
  - All powers are multiples of 3 so it must be a cube number
    - It can be written as $(2^3 \times 3 \times 5)^3$

How can I use PFD to find the square root of a square number?

- Write the number in its prime factor decomposition
  - All the indices should be **even** if it is a square number
- For example, to find the square root of 144 = $2^4 \times 3^2$
  - **Halve** all of the indices
    - $2^2 \times 3$
    - So $\sqrt{2^4 \times 3^2} = 2^2 \times 3$
- This is the prime factor decomposition of the **square root** of the number
  - To find it as an integer, multiply the prime factors together
  - $2^2 \times 3 = 12$, so the square root of 144 is 12

How can I use PFD to find the exact square root of a number?

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- $1440 = 2^5 \times 3^2 \times 5$

- Rewrite the prime factor decomposition with **as many even indices as you can**
  - E.g. $2^3 = 2^2 \times 2$, or $5^7 = 5^6 \times 5$
  - $1440 = 2^4 \times 2 \times 3^2 \times 5$

- Collect the terms with even powers together
  - $1440 = 2^4 \times 3^2 \times 2 \times 5$

- Square root both sides
  - $\sqrt{1440} = \sqrt{2^4 \times 3^2 \times 2 \times 5}$

- Using the rule $\sqrt{ab} = \sqrt{a}\sqrt{b}$, apply the square root to the terms with the even indices separately to the terms with odd indices
  - $\sqrt{1440} = \sqrt{2^4 \times 3^2} \times \sqrt{2 \times 5}$

- Simplify to find your answer, remembering that $\sqrt{a^2b^2} = ab$
  - $\sqrt{1440} = 2^2 \times 3 \times \sqrt{10}$
  - $\sqrt{1440} = 12\sqrt{10}$
  - $12\sqrt{10}$ is the **exact** square root of 1440

Worked Example

$N = 2^3 \times 3^2 \times 5^7$ and $AN = B$ where A is an integer and B is a non-zero square number.

Find the smallest value of A.

Answer:

Substitute $N = 2^3 \times 3^2 \times 5^7$ into the formula $AN = B$

$$A(2^3 \times 3^2 \times 5^7) = B$$

To be a square number, the prime factors of AN must all have even powers

Consider the prime factors A needs to have to make all the values on the left hand side have even powers

$$(2 \times 5)(2^3 \times 3^2 \times 5^7) = B$$

$$2^4 \times 3^2 \times 5^8 = B$$

So A, when written as a product of its prime factors, is 2×5

Make sure you write A as an integer value in the answer

$$A = 10$$

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